



Lab 02: Formatted I/O

CSE 4108

Structured Programming I Lab



Lab Tasks

1. Phone Number Formats:

Write a program that prompts the user to enter a telephone number in the form **(ddd)ddd-dddd** and then displays the number in the form **ddd.ddd.dddd**.

Sample Run:

```
Enter phone number [(ddd) ddd-dddd]: (404)
817-6900
You entered 404.817.6900
```

2. Inventory Management System:

Mr. Sakamoto from Convenience Delivery Store (CDS) of Imperial University of Tokyo (IUT) is back. This time, he wants you to create a list of items in his shop. He will provide you item number, unit price and purchase date of the products available in CDS. He wants you to show the items in an organized manner. However, before finalizing the design he wants to see how the list will look like a single item. You have to write a program that formats product information entered by the user. A session with the program should look like this:

Enter item number: 583

Enter unit price: 13.5

Enter purchase date (mm/dd/yyyy): 10/24/2019

Item	Unit Price	Purchase Date
583	13.50	10/24/2019

The item number and date should be left justified; the unit price should be right justified. Allow dollar amounts up to \$9999.99.

Hint: Use tabs to line up the columns

3. Numbers Square:

Write a program that asks the user to enter the numbers from 1 to 16 (in any order) and then displays the numbers in a 4 by 4 arrangement, followed by the sums of the rows, columns, and diagonals.

Sample Run:

Enter the numbers from 1 to 16 in any order:

16 3 2 13 5 10 11 8 9 6 7 12 4 15 14 1

Arranging as a square:

16	3	2	13
5	10	11	8
9	6	7	12
4	15	14	1

Row sums: 34 34 34 34
Column sums: 34 34 34 34
Diagonal sums: 34 34

**Note: The underlined string is the input.
Everything else is the printed output.**

If the row, column, and diagonal sums are all the same (as they are in this example), the numbers are said to form a magic square. The magic square shown here appears in a 1514 engraving by artist and mathematician Albrecht Dürer. (Note that the middle numbers in the last row gives the date of the engraving).

4. Treat! Treat!:

In 328-N, it's kind of a rule that whenever someone in the room gets GPA 4.00 in a semester, he has to give treat to the others. As always, Xedd got GPA 4.00 again and now its time for his treat. But this time Xedd wants to keep track of how much he has spent for Ferb, one of his roommates, as Ferb has no limits when it comes to eating. Xedd has decided that he'll not spend more than 100 taka for Ferb and he has allowed

Ferb to take at most 3 items in that price range. So as Ferb always wants to maximize the amount of food he's taken, he'll take all 3.

Write a C program that takes input the price of food items Ferb has taken in taka and prints out a receipt that contains the price and percentage of each individual item in the final bill along with the total price in taka.

Sample Input

Enter Khichuri price (taka): 68.68

Enter Doi price (taka): 20.15

Enter Drinks price (taka): 9.5

Sample Output

Final Bill -

Khichuri:	68.68tk	69.84%
Doi:	20.15tk	20.49%
Drinks:	9.50tk	9.66%
Total Price:	98.33tk	100.00%

Hint: Use tabs to line up the columns. Also notice that the price column is right aligned.

5. Fraction Subtraction:

Modify the addfrac.c program (shown in class) so that the user enters both fractions at the same time, separated by a minus sign.

Sample Run:

Enter two fractions separated by a minus sign:

5/6-3/4

The sum is 2/24

Hint: Fractional subtraction can be calculated using the following equation:

$$\frac{a}{b} - \frac{c}{d} = \frac{(a \times d) - (b \times c)}{b \times d}$$